Cody Pych

***COURSE*: CHE 312: Transport Phenomena II**

Fall Semester 2025

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Department of Chemical Engineering

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Quiz #4

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Open Notes and Book

Duration: 40 minutes

Consider the irreversible conversion of species A into B and F through the reactions A → B and A → F in a stationary liquid film of thickness L. Determine C_A throughout the film, if the surface at y = L is assumed to be impermeable and unreactive, and the concentration of A at y = 0 is specified as C_o. The homogeneous reactions follow zeroth and first order kinetics with rates of reaction $R_{AB} = -k_B$ and $R_{AF} = -k_F \cdot C_A$, respectively. The diffusivity of the species A is D_A.

Find the overall rate of reactant consumption.

$A \rightarrow B$ $\qquad$ $A \rightarrow F$

$R_{AB} = -K_B$ $\qquad$ $R_{AF} = -K_F C_A$

$R_A = -K_B - K_F C_A$

$C_A\big|_{y=0} = C_o$

$C_A\big|_{y=L} = 0$

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$$D_A \frac{d^2 C_A}{dy^2} - K_F C_A - K_B = 0$$

$$D_A \frac{d^2 C_A}{dy^2} - K_F \left(C_A + \frac{K_B}{K_F}\right) = 0$$

$$D_A \frac{d^2}{dy^2}\left(C_A + \frac{K_B}{K_F}\right) - K_F\left(C_A + \frac{K_B}{K_F}\right) = 0$$

$$C_A(y) = He^{\sqrt{\frac{K_F}{D_A}}\,y} + Je^{-\sqrt{\frac{K_F}{D_A}}\,y} - \frac{K_B}{K_F}$$

$$C_A(y=0) = H + J = C_0$$

$$C_A(y=L) = He^{\sqrt{\frac{K_F}{D_A}}\,L} + Je^{\sqrt{\frac{K_F}{D_A}}\,L}$$